Lakehead University
CS 1411 COMPUTER PROGRAMMING I
Practical Assignment No. 3

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Q1: A high school has 1000 students and 1000 lockers, one locker for each student. On the first day of school, the principle plays the following game: She asks the first student to go and open all the lockers. She then asks the second student to go and close all the even-numbered lockers. The third student is asked to check every third locker. If it is open, the student closes it; if it is closed, the student opens it. The fourth student is asked to check every fourth locker. If it is open, the student closes it; if it is closed, the student opens it. The remaining students continue this game. In general, the nth student checks every nth locker. If the locker is open, the student closes it; if it is closed, the student opens it. After all the students have taken their turn, some of the lockers are open and some are closed. Write a program that prompts the user to enter the number of lockers in a school. After the game is over, the program outputs the number of lockers that are opened **using nested for statement**. Test run your program for the following inputs: 1000, 5000, 10000. Do you see any pattern developing?
(Hint: Consider locker number 100. This locker is visited by student numbers 1, 2, 4, 5, 10, 20, 25, 50 and 100. These are the positive divisors of 100. Similarly, locker number 30 is visited by student number 1, 2, 3, 5, 6, 10, 15 and 30. Notice that if the number of positive divisors of a locker number is odd, then at the end of the game the locker is opened. If the number of positive divisors of a locker number is even, then at the end of the game the locker is closed.)

```
Enter the number of lockers:
10
Lockers that are opened:
1, 4, 9,
Number of opened lockers: 3
Press any key to continue . . .

Enter the number of lockers:
30
Lockers that are opened:
1, 4, 9, 16, 25,
Number of opened lockers: 5
Press any key to continue . . .

Enter the number of lockers:
100
Lockers that are opened:
1, 4, 9, 16, 25, 36, 49, 64, 81, 100,
Number of opened lockers: 10
Press any key to continue . . .

Enter the number of lockers:
1000
Lockers that are opened:
1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 36
1, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900, 961,
Number of opened lockers: 31
Press any key to continue . . .
```

Q2: When you borrow money to buy a house, a car, or for some other purpose, you repay the loan by making periodic payments over a certain time. Of course the landing company will charge interest on the loan. Every periodic payment consists of the interest on the loan and the payment toward the principle amount. To be specific, suppose that you borrow $1000at the interest rate of 7.2%per year and the payments are monthly. Suppose that your monthly payment is $25. Now the interest is 7.2% per year and the payments are monthly, so the interest rate per month is 7.2/12=0.6%. The first month's interest on $1000 is 1000*0.006=6. Because the payment is $25 and the interest for the first month is $6, the payment towards the principle amount is 25-6=19. This means after making the first payment, the loan amount is 1000-19=981. For the second payment, the interest is calculated on $981. So the interest for the second month is 981*0.006=5.886, that is, approximately $5.89. This implies that the payment towards the principal is 25-5.89=19.11 and the remaining balance after the second payment is 981-19.11=961.89. This process is repeated until the loan is paid. Write a program that accepts as inputs the loan amount, the interest rate per year, and the monthly payment. (Enter the interest rate as a percentage. For example, if the interest rate is 7.2%per year, then enter 7.2) Then program then outputs the number of months it would take to repay the loan. (Note that if the monthly payment is less than the first month's interest, then after each payment, the loan amount will increase. In this case, the program must warn the borrower that the monthly payment is too low and with this monthly payment the loan amount could not be repaid.)

Q3:

1- Write a program to print the following pattern using nested loops:

```
    *
   * *
  *   *
 *     *
*********
```

2- Write a program to print the following pattern using nested loops:

```
    A
   ABC
  ABCDE
 ABCDEFG
ABCDEFGHI
```

Q4: To maintain one's body weight, an adult human needs to consume enough calories daily to (1) meet the basal metabolic rate (energy required to breath, maintain body temperature, etc.), (2) account for physical activity such as exercise, and (3) account for the energy required to digest the food that is being eaten. For an adult that weighs P pounds we can estimate these caloric requirements using the following formulas:

1. Basal metabolic rate: calories required=70*(p/2.2)^0.756
2. Physical activity: calories required= 0.0385 * intensity * p * minutes

Here, minutes is the number of minutes spent during the physical activity, and intensity is a number that estimates the intensity of the activity. Here are some sample numbers for the range of values:

Activity	intensity
Running 10 mph:	17
Running 6 mph:	10
Basketball:	8
Walking 1 mph:	1

3. Energy to digest food: calories required=total calories consumed*0.1. In other words, 10% of the calories we consume goes towards digestion.
Write a function that computes the calories required for the basal metabolic rate, taking as input a parameter for the person's weight. **Write another function that** computes the calories required for physical activity, taking as input parameters for the intensity, weight, and minutes spent exercising.
Use these functions in a program that inputs a person's weight, an estimate for the intensity of physical activity, the number of minutes spent performing the physical activity, and the number of calories in one serving of your favorite food. The program should then calculate and output how many servings of that food should be eaten per day to maintain the person's current weight at the specified activity level. The computation should include the energy that is required to digest food. You can find estimates of the caloric content of many foods on the web. For example, a double cheeseburger has approximately 1000 calories.

Due on February 28